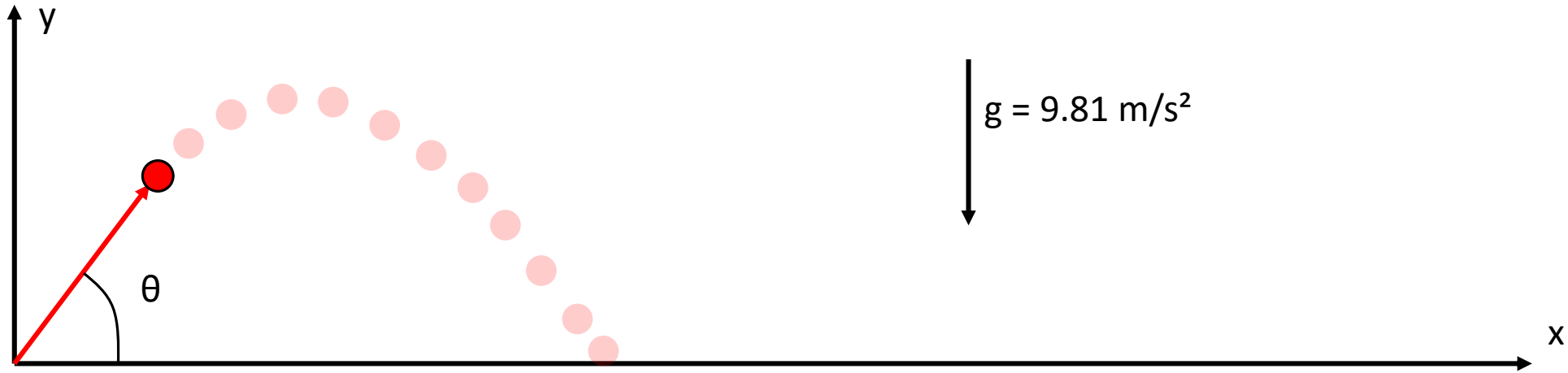
Simulation Multi-Physique (Application sous MATLAB & Simulink)

CONTACT : HUSSEIN SLEIMAN

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Example: simulating a projectile

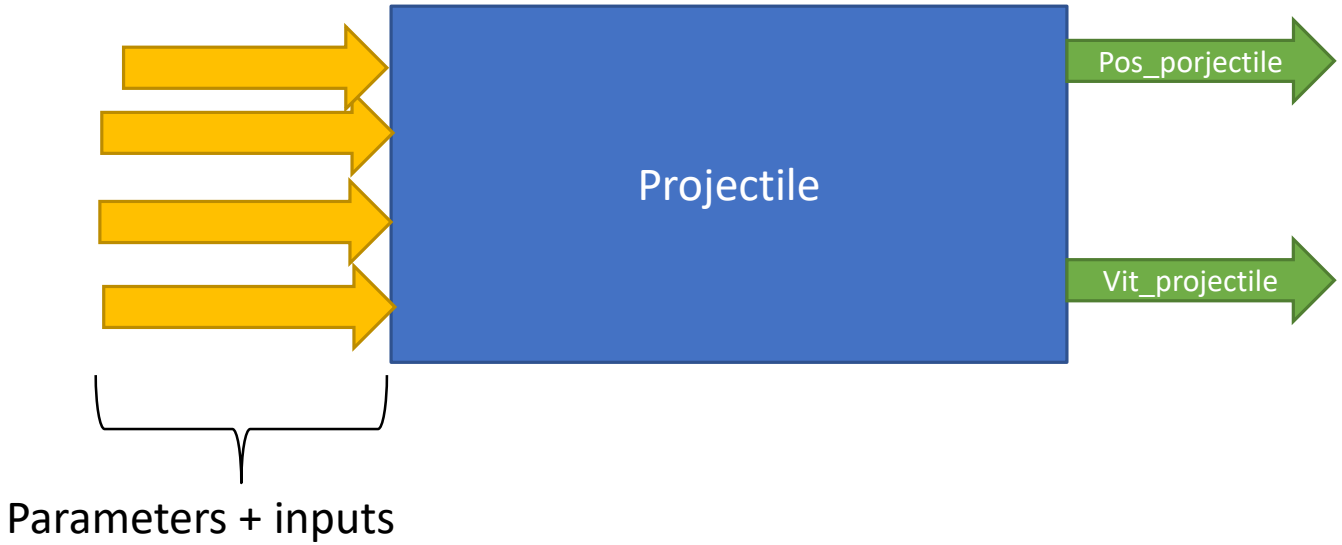


- **Write down the basic equation of the projectile's trajectory**
- **Model the problem in Simulink, simulate different cases**
- **Model the system more realistically, taking into account a non-zero initial position, non-zero initial speeds and considering air friction in the form of a force: F=-constant*V².**
- **Study the problem using MATLAB :**
  - **Parameter file**
  - **Simulation**
  - **Plotting results**

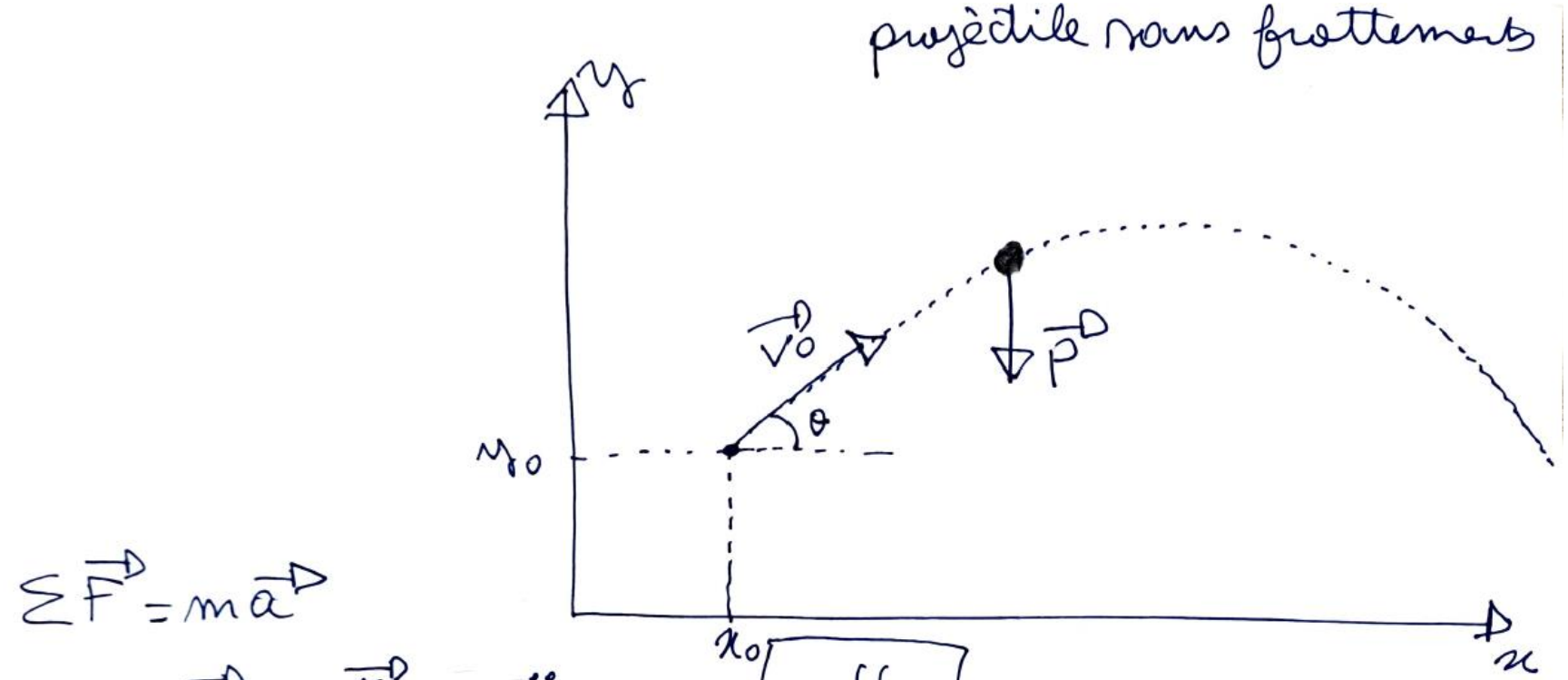
Example: simulating a projectile

Reminder: the system modeling process can be broken down into 6 stages:

- System definition: **projectile (bullet, shell, etc.)**
- Identifying variables: **positions and speeds**
- Parameters: **mass, gravity, air coefficient**



Example: simulating a projectile



$$\sum \vec{F} = m\vec{a}$$

$$\rightarrow x : \vec{0} = m\ddot{\vec{x}} \Rightarrow \ddot{x} = 0 \Rightarrow \boxed{x = \iint 0}$$

$$\rightarrow y : \vec{P} = m\ddot{\vec{y}} \Rightarrow -mg = m\ddot{y} \Rightarrow \boxed{y = \iint -g}$$

Example: simulating a projectile

- **Equation of motion using Newton's law :**
  - **On the x-axis :**

$$\frac{d^2x}{dt^2} = 0$$

  - **On the y-axis :**

$$\frac{d^2y}{dt^2} = -g$$

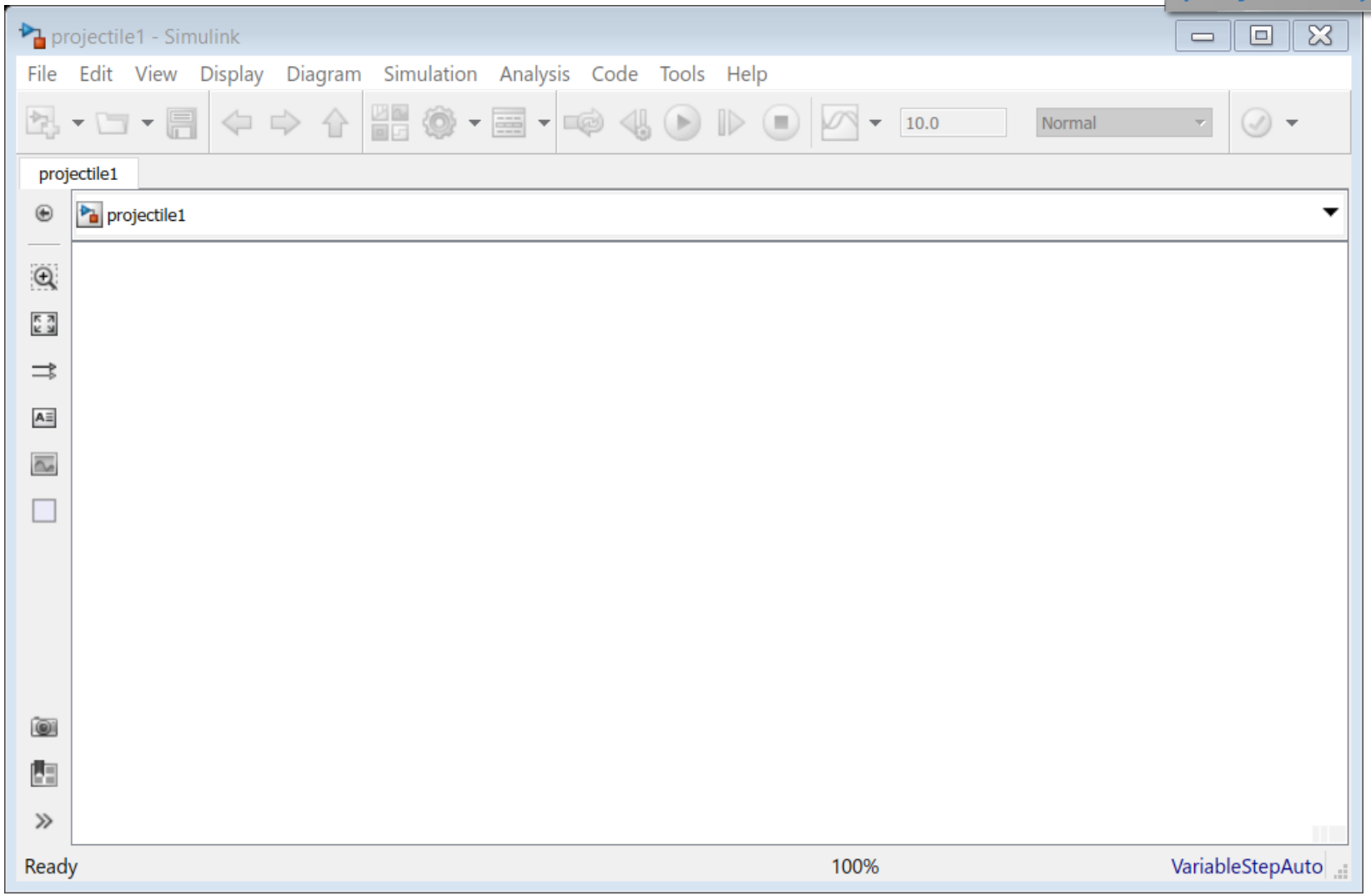
- **When modeling a physical problem, it is preferable to solve the problem by integration and not by derivation (causality problem):**

xpp(t) =cte1*t+cte2

$$\frac{d^2y}{dt^2} = -g \Rightarrow y = \int \left(\int -g dt \right) dt + cte2 \qquad (2)$$

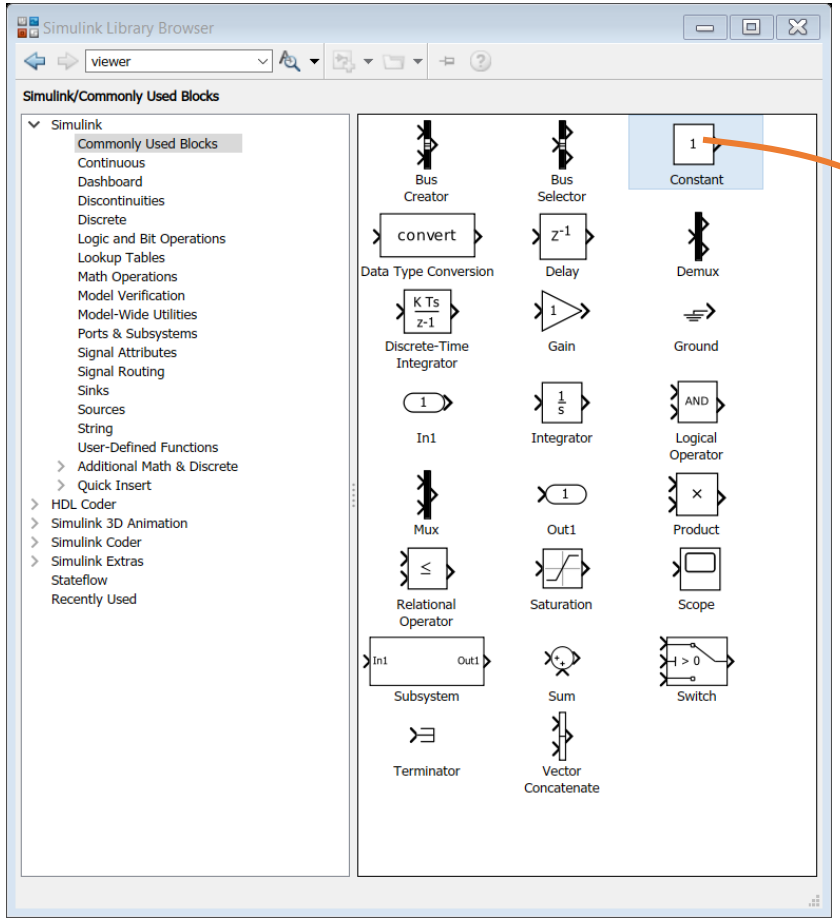
Example: simulating a projectile

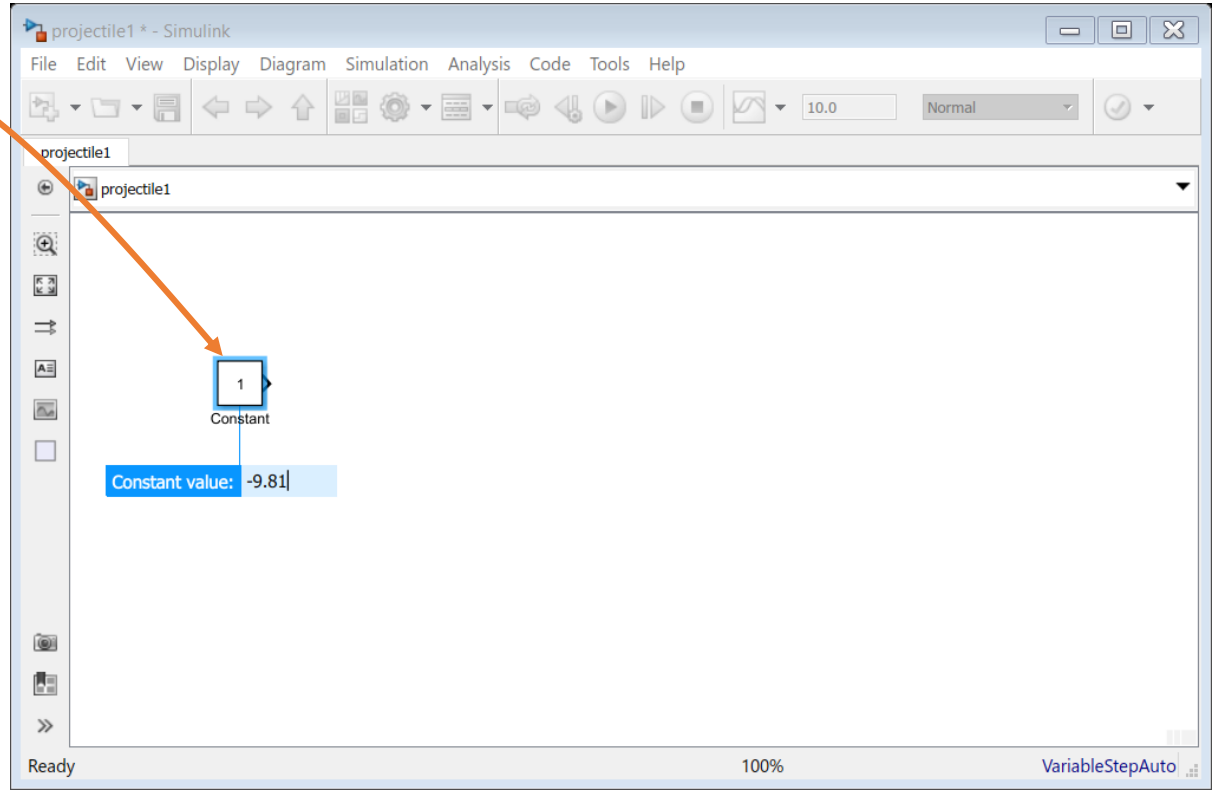
Save models and give them a name (e.g. projectile1)



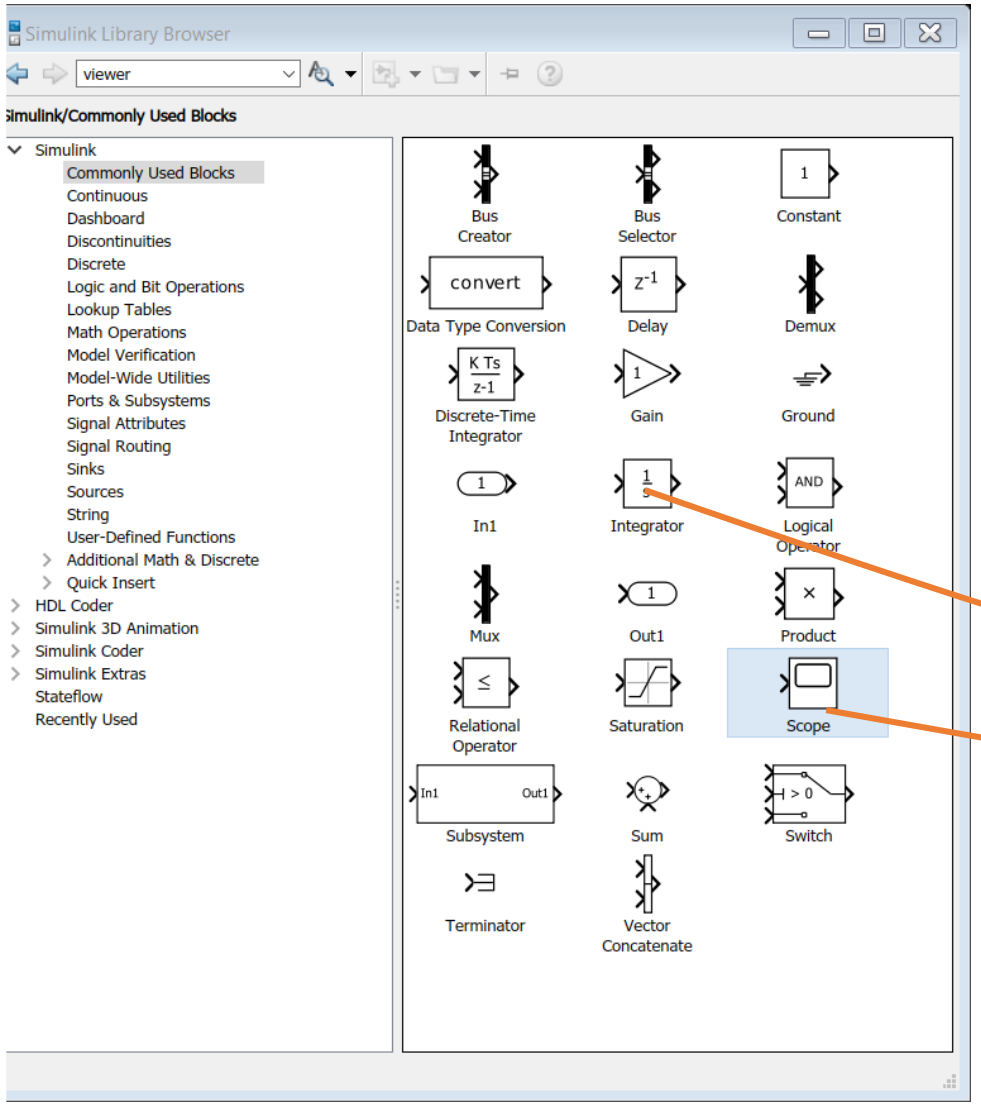
Example: simulating a projectile

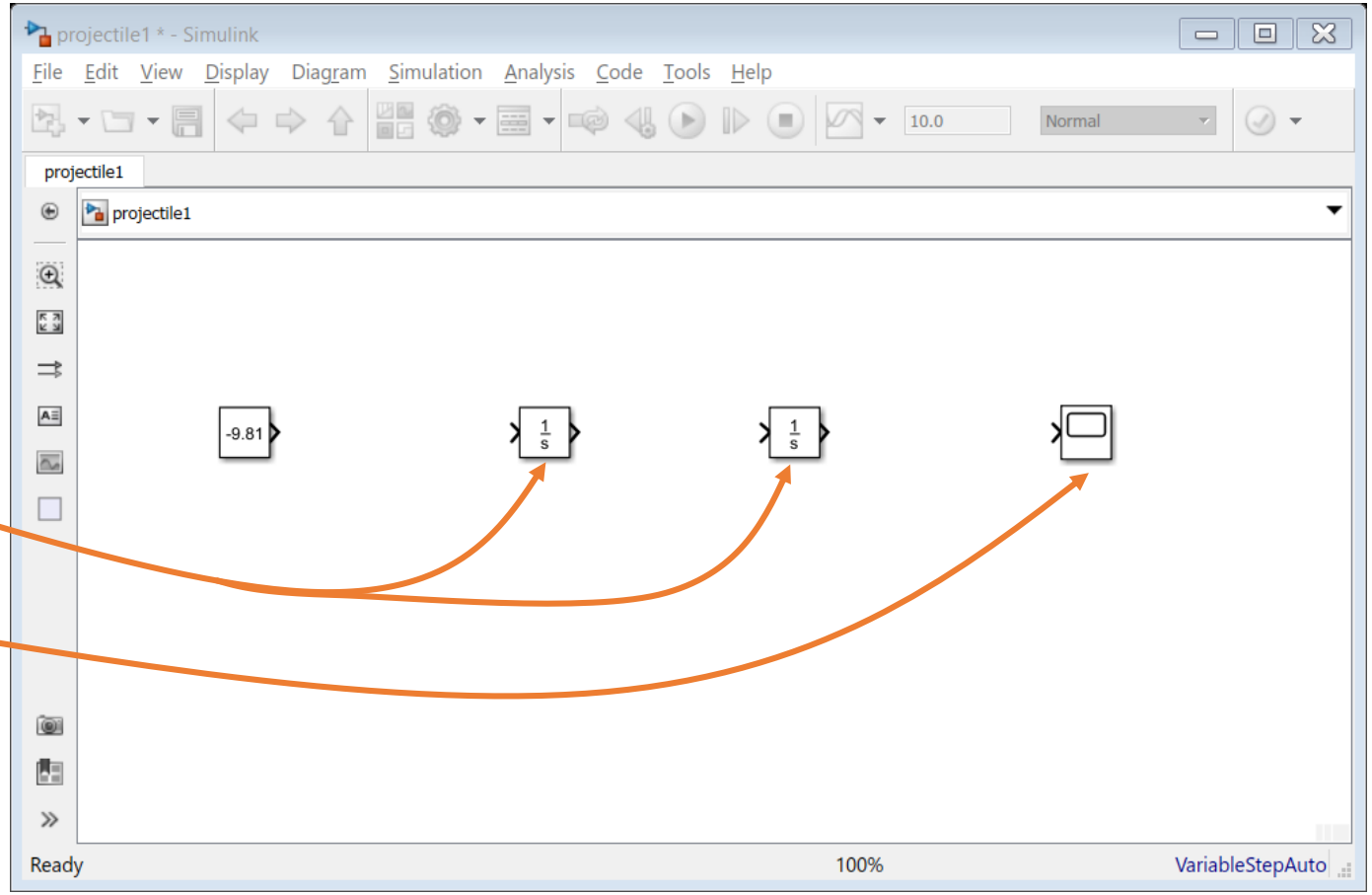
Drag and drop a constant to modify it.





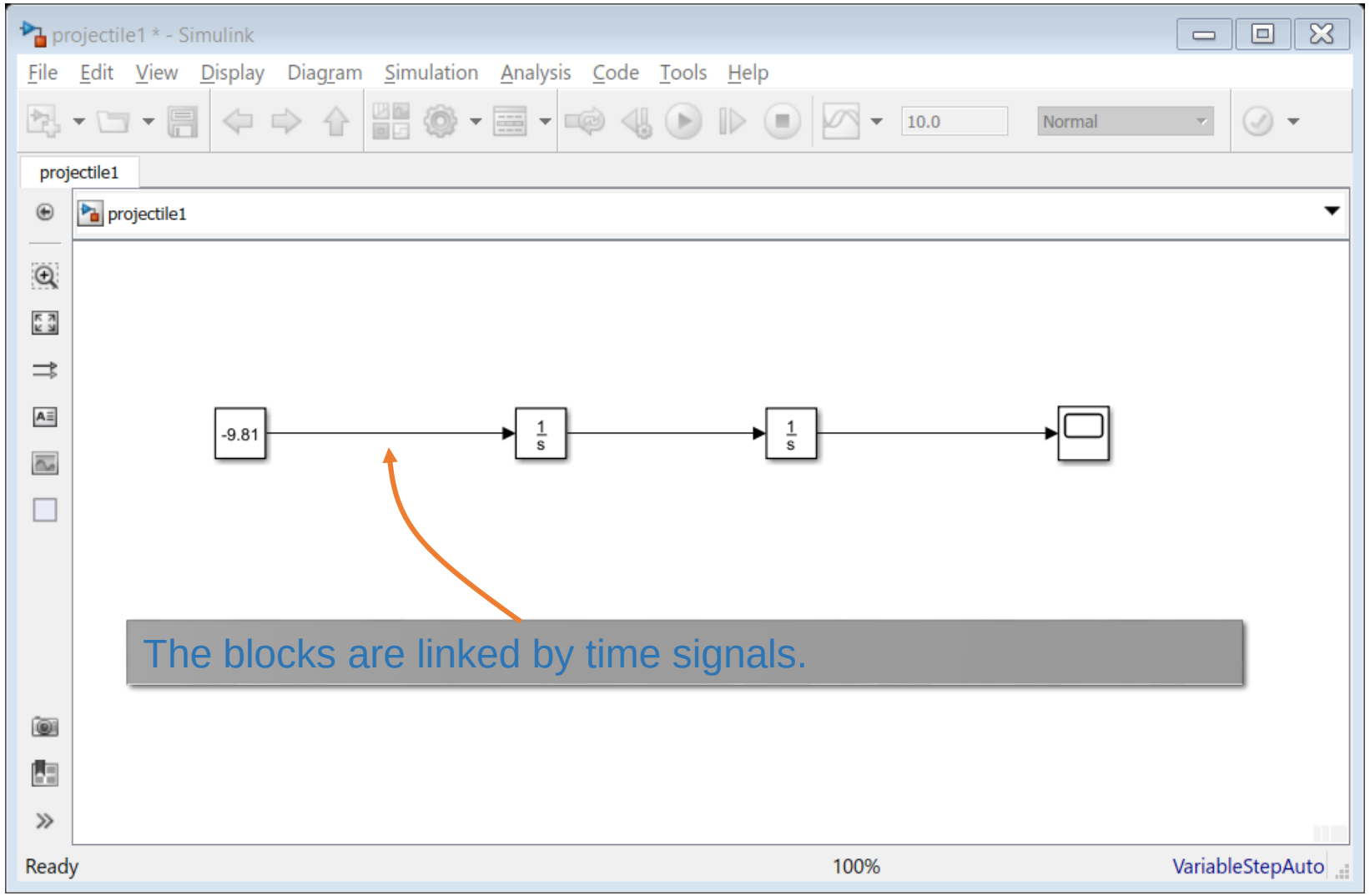
Example: simulating a projectile





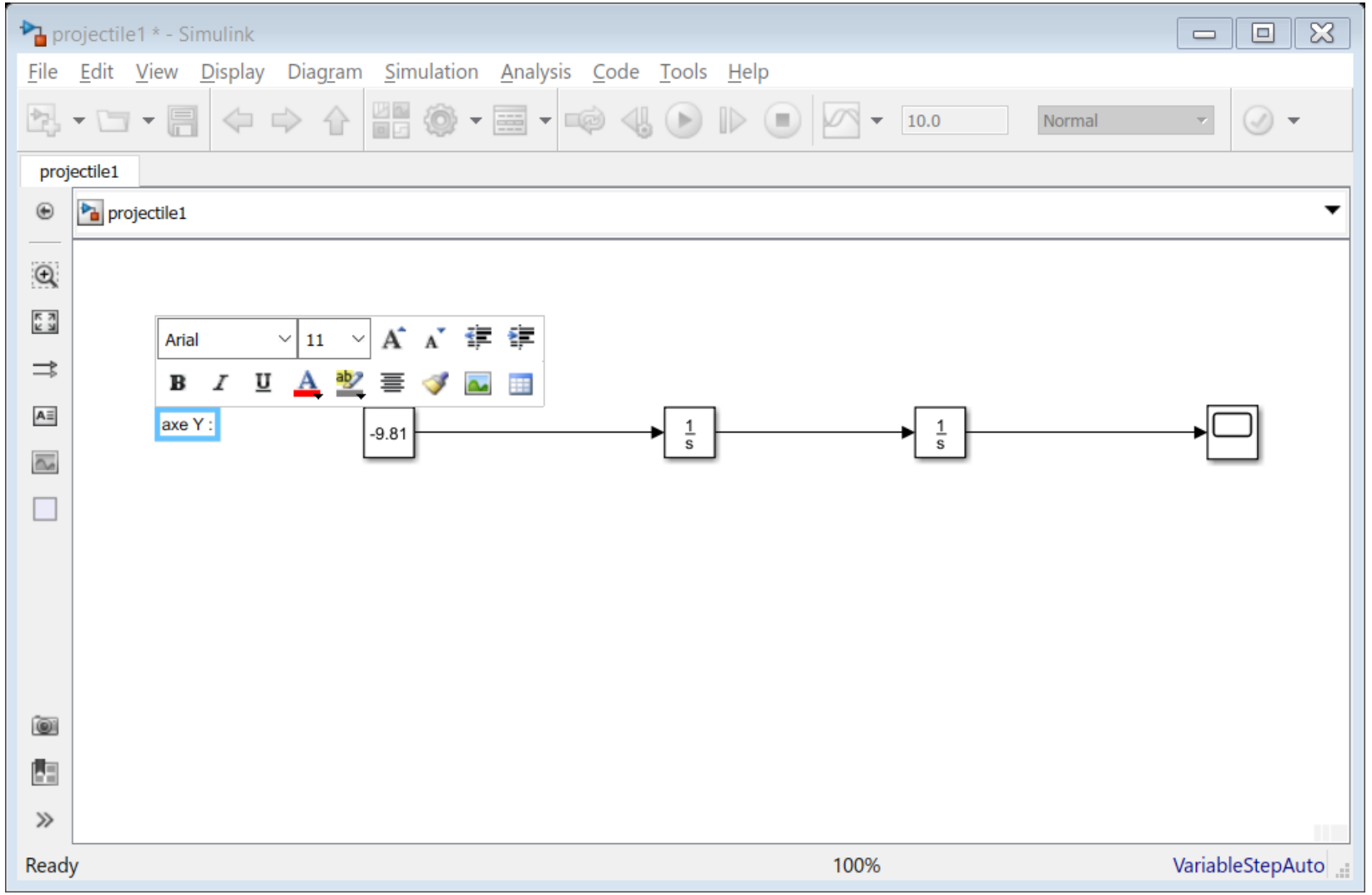
Example: simulating a projectile

To link blocks: click + *ctrl* key

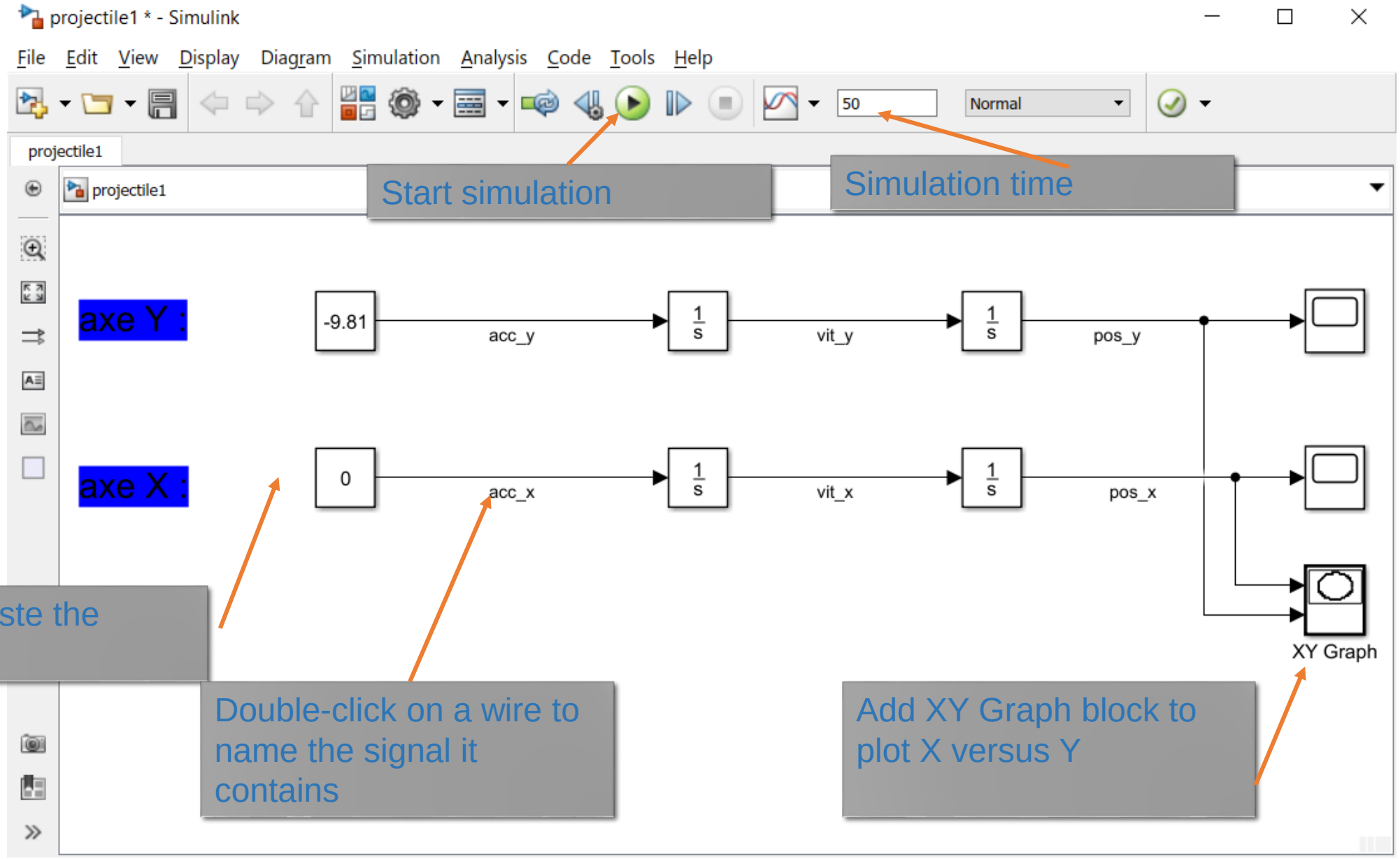


Example: simulating a projectile

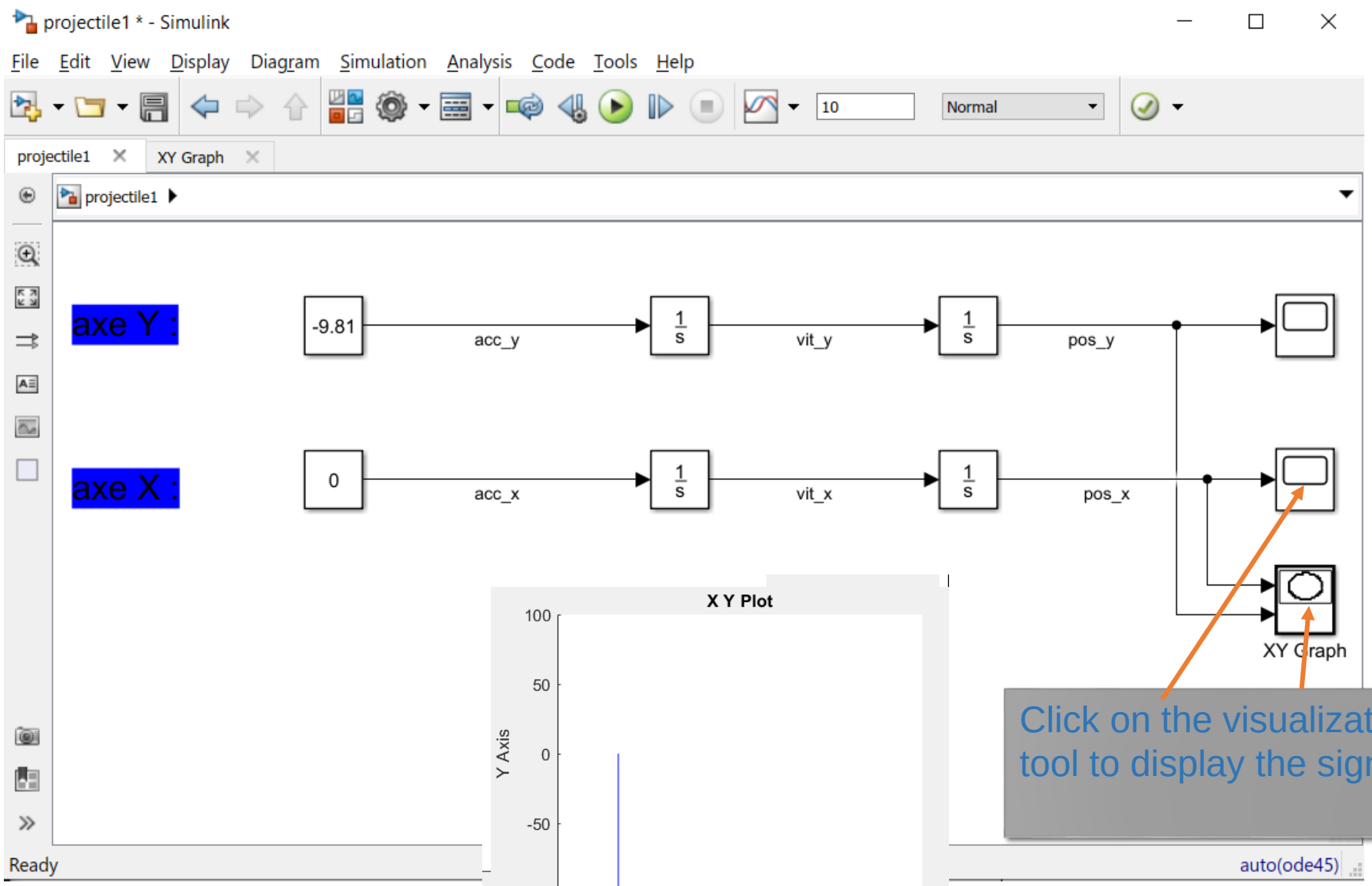
Add comments: double-click on an empty zone

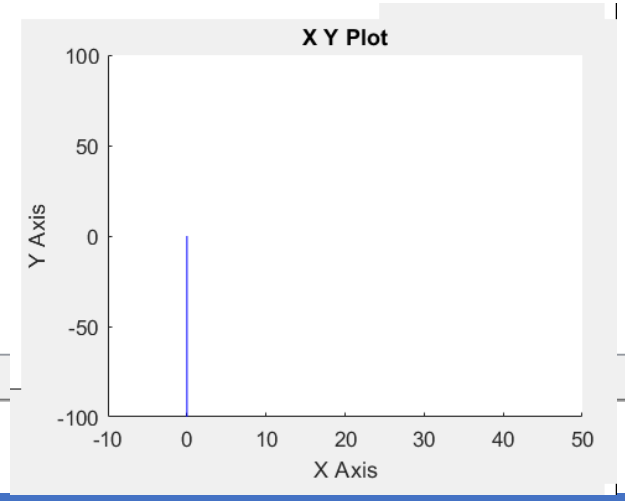


Example: simulating a projectile

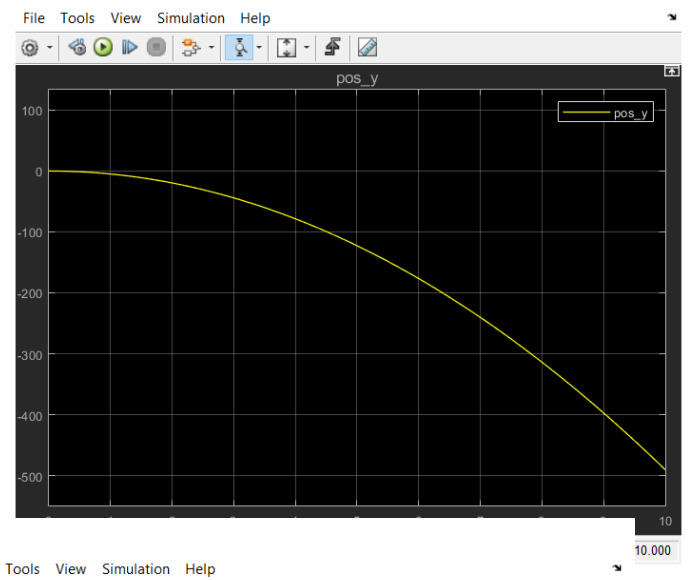


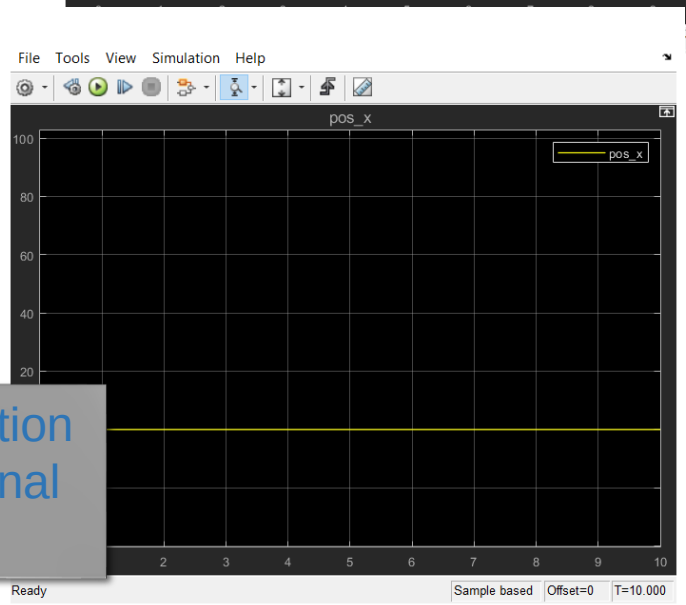
Example: simulating a projectile





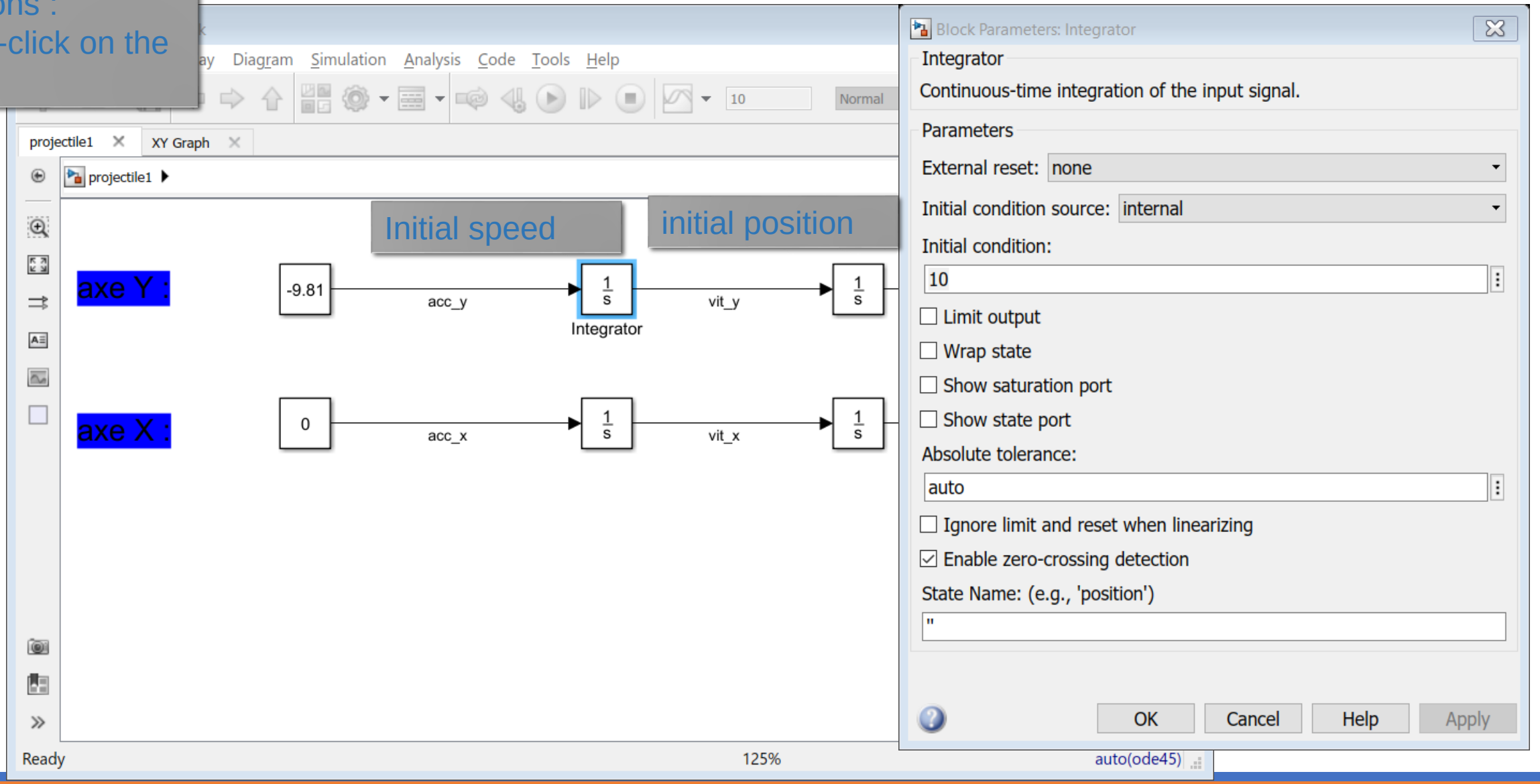
Click on the visualization tool to display the signal





Example: simulating a projectile

Modification of initial conditions : Double-click on the block

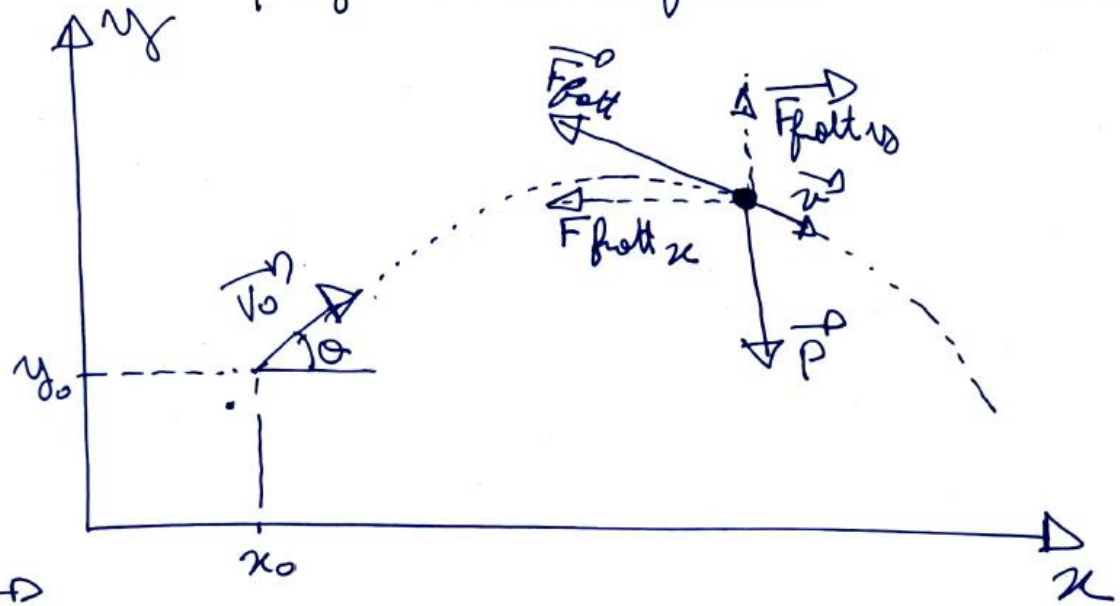


Example: simulating a projectile

- **Realistic modeling with air friction and non-zero initial conditions:**

projectile avec frottement de l'air

Frottement sous forme
$F_{frott} = -\alpha v^2$



$\sum \vec{F} = m\vec{a}$

$\rightarrow x : \vec{F}_{frott\,x} = m\vec{\ddot{x}} \Rightarrow -sign(\dot{x}) \cdot coeff_{frott} \cdot \dot{x}^2 = m\ddot{x} \Rightarrow$

$x = \iint \frac{1}{m}\left[-sign(\dot{x}) \cdot coeff_{frott} \cdot \dot{x}^2\right]$

$\rightarrow y : \vec{F}_{frott\,y} + \vec{P} = m\vec{\ddot{y}}$

$\Rightarrow -sign(\dot{y}) \cdot coeff_{frott} \cdot \dot{y}^2 - mg = m\ddot{y}$

$\Rightarrow y = \iint \frac{1}{m}\left[-sign(\dot{y}) \cdot coeff_{frott} \cdot \dot{y}^2 - mg\right]$

Example: simulating a projectile

MATLAB file

```
%% Date - Auteur
% ce fichier permet de :
% definir les paramètres du modele de projectile
% lancer la simulation
% tracer les resulat
clear all
clc
close all
```

```
%% paramètres physiques
masse           = 5      ; % (kg) masse du projectile
gravite         = 9.81   ; % (m/s²) acceleration de la gravité
coef_frottement = 0.02   ; % coef frottement de l'air
```

```
%% conditions initiales
vit0 = 100; %
theta = deg2rad(45) ;

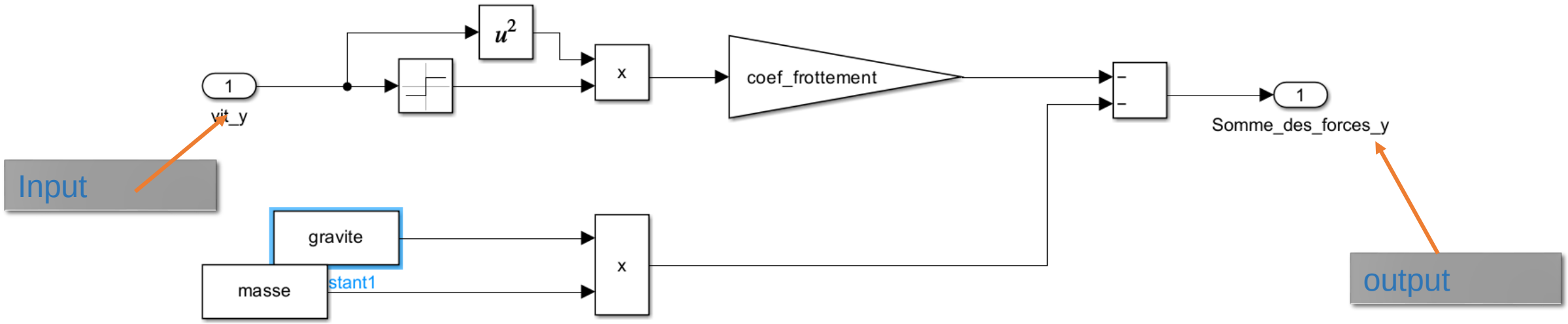
pos_x0 = 0 ; % (m) position initiale suivant x
vit_x0 = vit0*cos(theta) ; % (m/s) vitesse initiale suivant x

pos_y0 = 0 ; % (m) position initiale suivant y
vit_y0 = vit0*sin(theta) ; % (m/s) vitesse initiale suivant y
```

```
%% paramètres de simulation
duree_simu = 20 ; %(s) durée de simulation
```

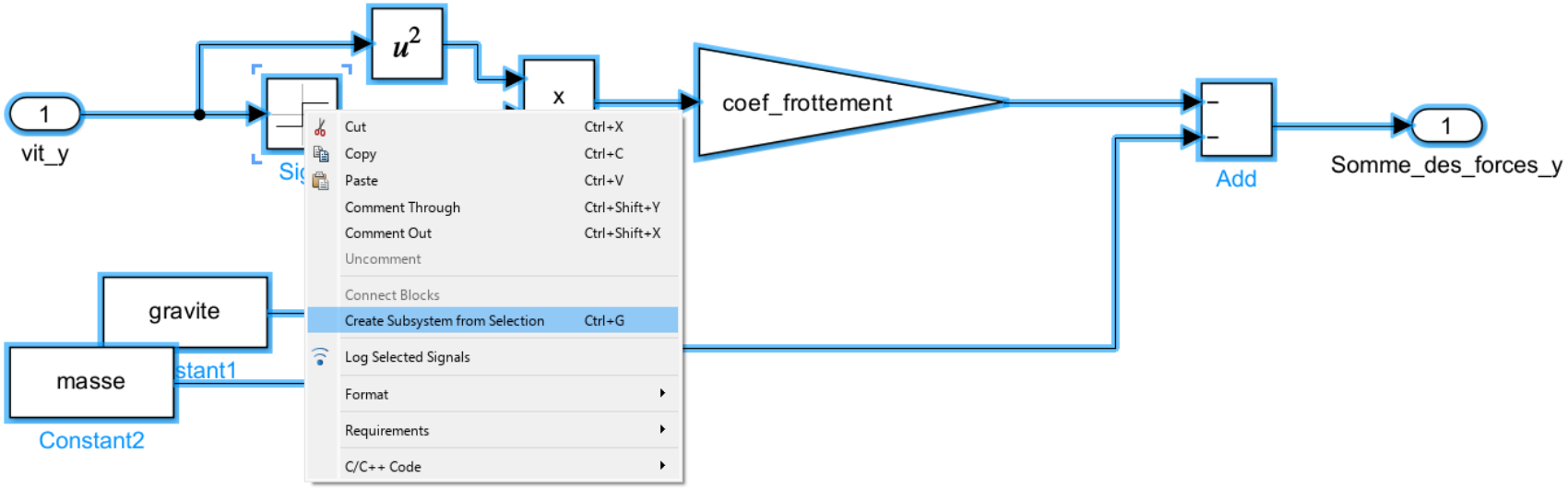
Example: simulating a projectile

Creation of a subsystem for forces applied to the projectile

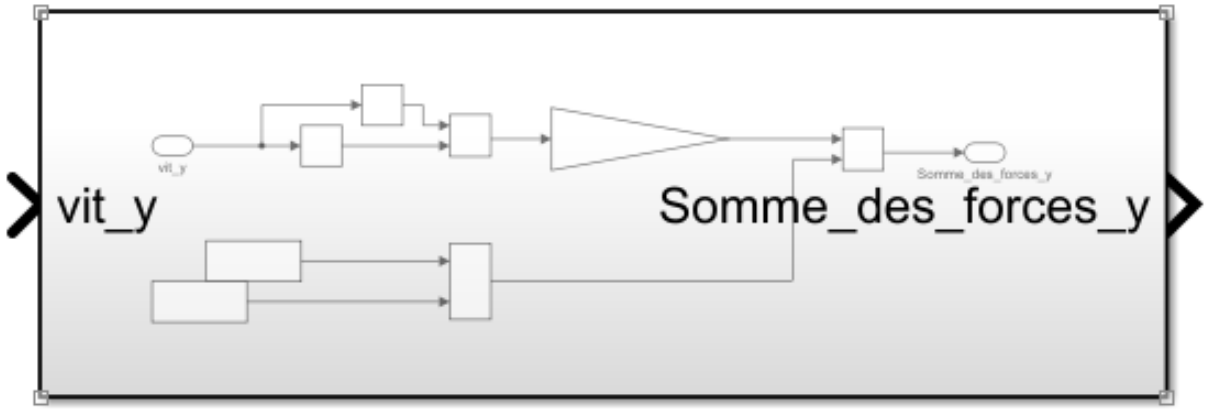


Example: simulating a projectile

Select all blocks
Right-click and create a subsystem

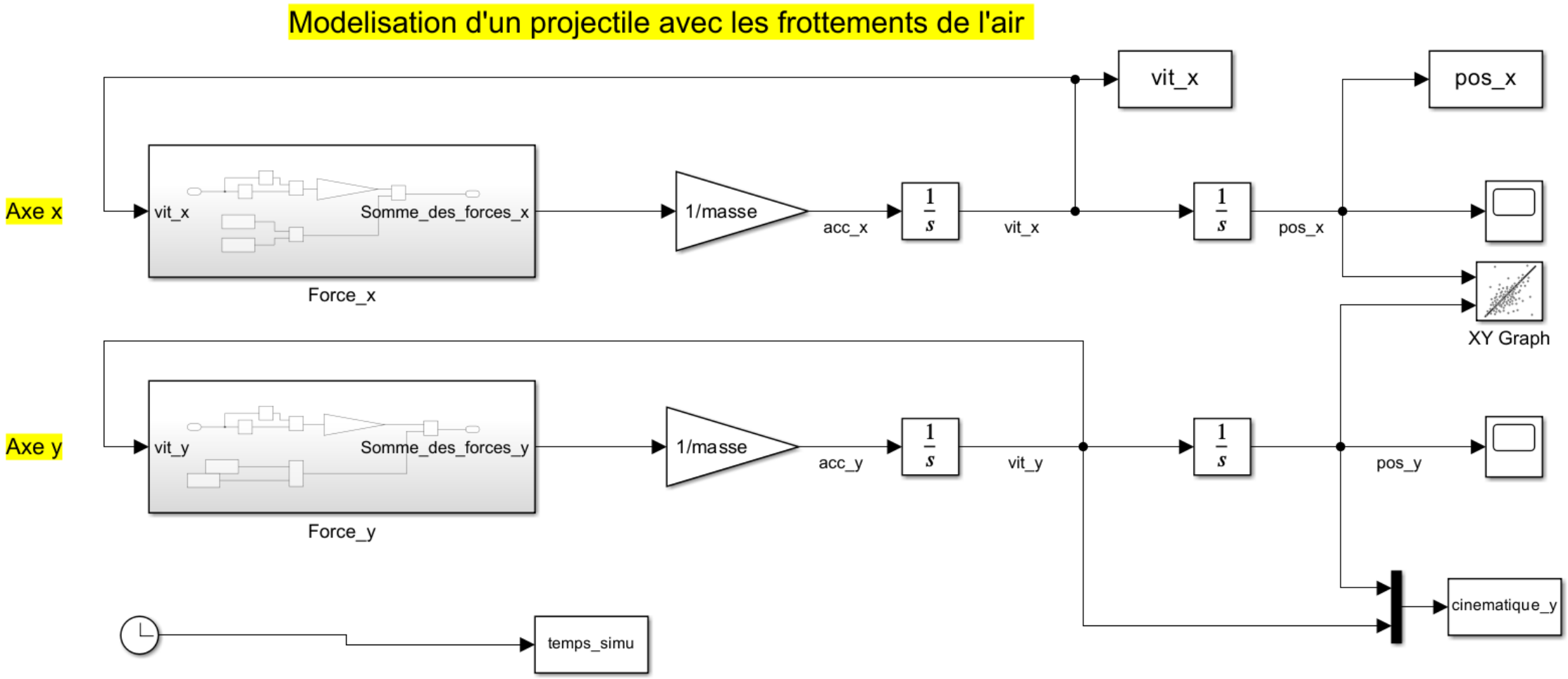


Example: simulating a projectile



Example: simulating a projectile

Modeling the two axes



Example: simulating a projectile

Modeling the two axes

